STAT lecture 9
Linear Regression: models

Disclaimer:

Please note that these slides are only a brief overview of the topic. Many more details will be explained on the blackboard and with MATLAB examples

SDU

Simple Linear Regression (analysis of association between variables)

Data — **Pairwise data** $(x_i, Y_i), \quad i = 1, \ldots, n$ where
x_i are deterministic (set) values for the predictor variable
Y_i are random (measured) values for the response variable

SLR model — $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \ldots, n$
β_0, β_1 are the unknown regression coefficients for intercept and slope
ε_i are unknown and random error terms for non-perfect linear model

Assumptions — $E[\varepsilon_i] = 0 \quad \rightarrow \quad E[Y_i] = \beta_0 + \beta_1 x_i$
$V[\varepsilon_i] = \sigma^2 \quad \rightarrow \quad V[Y_i] = \sigma^2$ (σ^2 constant but unknown)
No distribution assumptions (so far)

First tasks — **Estimate the unknown parameters** β_0, β_1 (to define regression line) and σ^2 (for inference methods)

Least Squares estimation of regression coefficients

Estimated regression $\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$

Residuals $\hat{\varepsilon}_i = Y_i - \widehat{Y}_i = Y_i - \left(\widehat{\beta}_0 + \widehat{\beta}_1 x_i\right)$

LS cost function Find $\widehat{\beta}_0, \widehat{\beta}_1$ to minimize $SSE = \sum_{i=1}^{n} \hat{\varepsilon}_i^{\,2} = \sum_{i=1}^{n} \left(Y_i - \left(\widehat{\beta}_0 + \widehat{\beta}_1 x_i\right)\right)^2$

$\nabla SSE\left(\widehat{\beta}_0, \widehat{\beta}_1\right) = 0$ gives (details on blackboard)

$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{x}$$

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(Y_i - \overline{Y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2} = \frac{(n-1)S_{xY}}{(n-1)S_{xx}} = \frac{S_{xY}}{S_{xx}}$$

Further

$$\widehat{\sigma}^2 = \frac{SSE}{df} = \frac{\sum_{i=1}^{n} \hat{\varepsilon}_i^{\,2}}{n-2} = \frac{\sum_{i=1}^{n}\left(Y_i - \left(\widehat{\beta}_0 + \widehat{\beta}_1 x_i\right)\right)^2}{n-2}$$

Model check Residual plots

Measure of model fit (details on blackboard)

$$R^2 \overset{\text{def}}{=} \frac{SSR}{SST} = \frac{\sum_{i=1}^{n}\left(\widehat{Y}_i - \overline{Y}\right)^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - \widehat{Y}_i\right)^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2}, \qquad 0 \le R^2 \le 1$$

What does ”linear” regression mean ?

Linear regression means that the regression model must be linear in the unknown regression coefficients, but not necessarily in x

E.g. The model $Y_i = \beta_0 + \beta_1 x_i^2 + \varepsilon_i$
is not linear in x_i (does not produce a straight line) but is linear in β_0, β_1

Define $z_i \stackrel{\text{def}}{=} x_i^2$ and the model becomes $Y_i = \beta_0 + \beta_1 z_i + \varepsilon_i$

Thus the linear regression technique can be used for the data set (z_i, Y_i) producing for this example the best fitting parabola to the data (x_i, Y_i)

Non-linear regression

Some – but not all - regression models that are not linear in the regression coefficients can be transformed into a linear model by transformation of variables

E.g The model $Y_i = \beta_0 \, exp(\beta_1 x_i)$ can be transformed into a linear model
(details on blackboard)